



Lesson 8.2 — Normal Distributions

Big idea: Many real-world quantities (heights, test scores, measurement errors) cluster around a center and taper off symmetrically — a normal distribution. The 68–95–99.7 rule lets you reason about it without computation.

Quick review

Shape: symmetric, bell-shaped, characterized by mean μ and standard deviation σ .

Empirical (68–95–99.7) rule: in a normal distribution, $\approx 68\%$ of data lies within 1σ of μ , $\approx 95\%$ within 2σ , $\approx 99.7\%$ within 3σ .

z-score: $z = (x - \mu)/\sigma$ tells you how many standard deviations x is above (or below) the mean.

Standard normal: mean 0, standard deviation 1. Use z-tables or technology to find areas (probabilities) under it.

Worked example

Worked example. SAT math scores are approximately normal with $\mu = 510$, $\sigma = 100$. Estimate the percent of students scoring between 410 and 610.

That's $\mu \pm \sigma \rightarrow$ about 68%.

Warm-ups

1. About what percent of a normal distribution is within 2σ of the mean?
2. Find the z-score of $x = 80$ if $\mu = 70$, $\sigma = 5$.
3. If $z = -1$, how many standard deviations is x below the mean?

Practice

1. Heights of women are approximately normal with $\mu = 65$ in, $\sigma = 3$ in. About what percent are between 59 and 71 in?
2. An IQ test has $\mu = 100$, $\sigma = 15$. About what percent of IQs are above 130?
3. Find the z-score of $x = 92$ if $\mu = 100$, $\sigma = 12$.

Challenge

1. Test scores are normally distributed with $\mu = 75$, $\sigma = 10$. (a) Approximate the probability a randomly chosen student scores above 95. (b) Approximate the probability of scoring between 65 and 85. (c) Find the z-score of a 90.
2. On a standardized test, scores are normal with $\mu = 500$, $\sigma = 100$. The top 16% of scores qualify for a program. Estimate the cutoff score.



Answer key — Lesson 8.2

Check your work. If a problem tripped you up, rewatch the step in the video where that concept appears.

Warm-ups

1. About 95%.
2. $z = (80 - 70)/5 = 2$.
3. 1 standard deviation below the mean.

Practice

1. 59 to 71 is $\mu \pm 2\sigma \rightarrow$ about 95%.
2. 130 is 2σ above the mean. Above $2\sigma \approx (100\% - 95\%)/2 = 2.5\%$.
3. $z = (92 - 100)/12 = -2/3 \approx -0.67$.

Challenge

1. (a) 95 is 2σ above mean. Above $2\sigma \approx 2.5\%$. (b) 65 to 85 is $\mu \pm \sigma \approx 68\%$. (c) $z = (90 - 75)/10 = 1.5$.
2. 16% is roughly the area above $z = 1$ (since 84% is below $z = 1$ by the empirical rule). Cutoff $\approx \mu + 1\sigma = 600$.